

MATH3026/4022: Time Series Analysis/Time Series and Forecasting
Exercises Sheet 3

To be covered during the examples class at 16:00 on Monday 9th March. These exercises do not count toward your final module mark, however it is a good idea to attempt these questions prior to the examples class.

1. Consider the process

$$X_t = Z_t + \frac{1}{4}Z_{t-1} - \frac{1}{8}Z_{t-2}$$

where $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with variance σ_z^2 .

- (a) Is the process $\{X_t\}$ invertible? Justify your answer.
- (b) Derive the autocovariance generating function (AGF) of this process.
- (c) Using your answer to (b), determine the autocorrelation function of the process $\{X_t\}$.

2. The process $\{Y_t : t \in \mathbb{Z}\}$ is defined

$$Y_t = \beta_0 + \beta_1 t + X_t$$

where $X_t = Z_t + 0.6Z_{t-1}$ and $\{Z_t\}$ is a white noise process with variance σ_z^2 .

- (a) Determine $\mathbb{E}(Y_t)$.
- (b) Determine the autocovariance function, $\gamma_Y(h)$, for the process $\{Y_t\}$.
- (c) The process $\{W_t\}$ is defined

$$W_t = Y_t - Y_{t-1} - \beta_1.$$

What type of process is $\{W_t\}$? Is this process stationary and/or invertible?

3. A sequential sample of 100 observations is made from a time series. The sample autocorrelations r_j ($j = 1, \dots, 10$), calculated using this sample of data, are shown in the table below

j	1	2	3	4	5	6	7	8	9	10
r_j	0.31	0.37	-0.05	0.06	0.22	0.11	0.08	0.05	0.12	0.01

Can you suggest a plausible *ARMA* process from which this sample of data may have arisen? Explain your answer.

4. Suppose that the time series generated by the $MA(1)$ process

$$X_t = (1 - \theta B)Z_t \quad (1)$$

is incorrectly identified as the $AR(1)$ process

$$(1 - \phi B)X_t = W_t \quad (2)$$

where $\{Z_t\}$ and $\{W_t\}$ are independent white noise processes, each with variance σ^2 . Assume that the model (2) is erroneously fitted to data generated by model (1). Discuss the form of the model residuals.

5. The model

$$X_t = (1 - 0.6B)Z_t$$

is fitted to a sample of time series data, where $\{Z_t\}$ is a white noise process with variance σ_z^2 . The residuals from the model appear to look like an $AR(1)$ model of the form

$$(1 - 0.4B)Y_t = W_t$$

where $\{W_t\}$ is a white noise process. What type of model would you suggest should be fitted next? Justify your answer.

6. Suppose that $\{X_t\}$ is a weakly stationary process with autocovariance function $\gamma_X(h)$. The sample mean of n successive observations of the process is defined

$$\bar{X}_n = \frac{1}{n} \sum_{t=1}^n X_t.$$

Determine the variance of $\bar{X}_n$, expressing your answer in as simple a form as possible.

7. Consider the $AR(2)$ process

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + Z_t$$

where $\{Z_t\}$ is a white noise process with variance σ_z^2 . Suppose that $\hat{\boldsymbol{\phi}} = (\hat{\phi}_1, \hat{\phi}_2)^\top$ denote estimators of the parameters $\boldsymbol{\phi} = (\phi_1, \phi_2)^\top$ formed by solving the Yule-Walker equations where a sample of size n of $\{X_t\}$ is taken.

Assuming that the $AR(2)$ process is (weakly) stationary, prove that

$$\text{Var} \left(n^{\frac{1}{2}} \left(\hat{\phi}_j - \phi_j \right) \right) \approx 1 - \phi_2^2 \quad \text{for } j \in \{1, 2\}$$

when n is large.

NOTE: You may assume that

$$n^{\frac{1}{2}} \left(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi} \right) \xrightarrow{D} \mathcal{N}_2 \left(\mathbf{0}, (1 - \boldsymbol{\rho}_2^\top \boldsymbol{\phi}) \mathbf{P}_2^{-1} \right)$$

where

$$\boldsymbol{\rho}_2 = (\rho_1, \rho_2)^\top; \quad \mathbf{P}_2 = \begin{bmatrix} 1 & \rho_1 \\ \rho_1 & 1 \end{bmatrix},$$

as $n \rightarrow \infty$.

8. Suppose $\{X_t\}$ is a (weakly) stationary process with mean zero and we define

$$D_m = \mathbb{E} \left[(X_t - \mathbf{X}_m^\top \boldsymbol{\beta}_m)^2 \right]$$

where

$$\begin{aligned} \mathbf{X}_m &= (X_{t-1}, \dots, X_{t-m})^\top; \\ \boldsymbol{\beta}_m &= (\beta_{m,1}, \dots, \beta_{m,m})^\top. \end{aligned}$$

- (a) By setting the derivatives of D_m with respect to $\beta_{m,1}, \dots, \beta_{m,m}$ to zero, show that finding $\boldsymbol{\beta}_m$ to minimise D_m is equivalent to finding $\boldsymbol{\phi} = (\phi_{m1}, \dots, \phi_{mm})^\top$ to solve the $m \times m$ system of Yule-Walker equations

$$\mathbf{P}_m \boldsymbol{\phi}_m = \boldsymbol{\rho}_m;$$

in the sense that $\boldsymbol{\beta}_m = \boldsymbol{\phi}_m$.

Here $\mathbf{P}_m$ is the $m \times m$ matrix whose $(i, j)^{\text{th}}$ element is ρ_{i-j} and $\boldsymbol{\rho} = (\rho_1, \dots, \rho_m)^\top$ with $\rho_h = \text{Cor}(X_t, X_{t+h})$.

- (b) Hence deduce that $\beta_{m,m}$, the partial autocorrelation function at lag m , is equal to zero for an $AR(p)$ model if $m > p$.

9. The dataset `USAirPassengers.csv` can be found on the course moodle page and contains monthly US Air Passenger numbers from 1949 to 1960, covered during Lecture 10. Using R, fit an appropriate time series model to these data and check the fit of your model.